

作业纸

课程名称: _____

班级: _____

教学班级: _____

姓名: _____

学号: _____

第 _____ 页

2.4.

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$n < 7$, $x[n]$ 与 $h[n-k]$ 无交点, 故 $y[n] = 0$

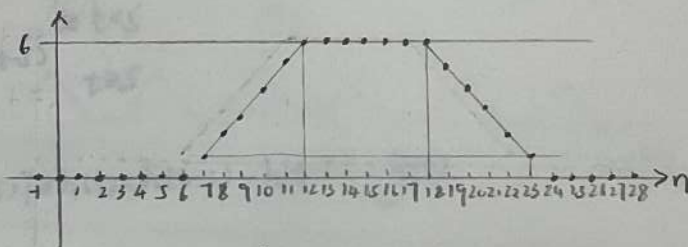
$$7 \leq n < 12, y[n] = n - 4 - 3 + 1 = n - 6$$

$$12 \leq n < 18, y[n] = 6$$

$$18 \leq n \leq 23, y[n] = 6 - (n - 18) = 24 - n$$

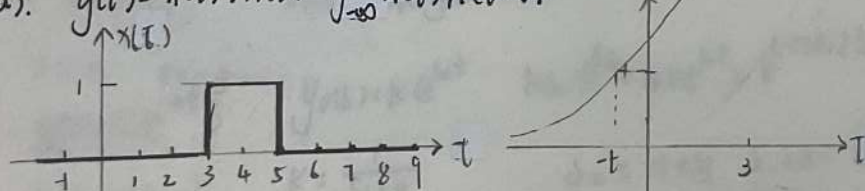
$$23 < n, y[n] = 0$$

$$y[n] = \begin{cases} n-6, & 7 \leq n < 12 \\ 6, & 12 \leq n < 18 \\ 24-n, & 18 \leq n \leq 23 \\ 0, & \text{其他} \end{cases}$$



2.11. $x(t) = u(t-3) - u(t-5)$ 且 $h(t) = e^{-3t}u(t)$

(a). $y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$



$t < 0$ 时, $y(t) = 0$

$0 < t < 3$ 时, $y(t) = 0$

$3 \leq t \leq 5$ 时, $y(t) = \int_3^5 h(t-\tau)d\tau = \int_3^5 e^{-3(t-\tau)}d\tau = \int_0^{5-t} e^{-3\tau}d\tau = \int_0^{5-t} \frac{1}{3}e^{-3\tau}d(3\tau) = \frac{1}{3}(1 - e^{-3(5-t)}) = \frac{1-e^{-3(5-t)}}{3}$

$t \geq 5$ 时, $y(t) = \int_3^5 e^{-3(t-\tau)}d\tau = \frac{1}{3}[e^{-3(5-t)} - e^{-3(3-t)}] = \frac{1}{3}e^{-3t}[e^{15} - e^9]$

$$y(t) = \begin{cases} 0 & -\infty < t < 3 \\ \frac{1-e^{-3(5-t)}}{3} & 3 \leq t \leq 5 \\ \frac{(1-e^{-3t})e^{-3(5-t)}}{3} & 5 < t < \infty \end{cases}$$

联系方式: _____

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课程名称: _____

班级: _____

教学班级: _____

姓名: _____

学号: _____

第 _____ 页

(b) $g(t) = (dx(t)/dt) * h(t)$

$$dx(t)/dt = \delta(t-3) - \delta(t-5)$$

$$g(t) = [\delta(t-3) - \delta(t-5)] * h(t) = h(t-3) - h(t-5) = e^{-3(t-3)}u(t-3) - e^{-3(t-5)}u(t-5)$$

$$= \begin{cases} 0 & t < 3 \\ e^{-3(t-3)} & 3 \leq t < 5 \\ e^{-3(t-3)} - e^{-3(t-5)} & t \geq 5 \end{cases}$$

(c) $\frac{dy(t)}{dt} = [\delta(t-3) - \delta(t-5)] * h(t) = h(t-3) - h(t-5) = g(t)$

2.17. $\frac{d}{dt}y(t) + 4y(t) = x(t)$

(a) $x(t) = e^{(-1+2j)t}u(t)$. 求 $y(t)$

$r+4, r=-4, y_p(t) = Ke^{s_0 t} = Ke^{(-1+2j)t}$

$\therefore y_h(t) = Ce^{-4t}$ $K(s_0+4) = 1 \quad K = \frac{1}{s_0+4} \quad s_0 = -1+2j \quad s_0+4 = 3(1+j) \Rightarrow K = \frac{1}{3} \frac{1}{1+j} = \frac{1-j}{6}$

$y(t) = y_h(t) + y_p(t) = Ce^{-4t} + \frac{1}{3\sqrt{2}} e^{-j\frac{\pi}{4}} e^{s_0 t} = \frac{1}{3\sqrt{2}} e^{s_0 t - j\frac{\pi}{4}} + Ce^{-4t}$

$y(0) = 0 \quad C + \frac{1}{3\sqrt{2}} e^{-j\frac{\pi}{4}} = C + \frac{1}{3(1+j)} = 0 \quad C = -\frac{1}{3(1+j)}$

$y(t) = \frac{e^{-4t}}{3\sqrt{2}} [e^{(-1+2j)t} - e^{-4t}] u(t)$

(2) $\text{Re}y(t) = \frac{e^{-4t}}{3\sqrt{2}} \cos(2t - \frac{\pi}{4}) \quad \text{Re}y_2(t) = \frac{e^{-4t}}{3\sqrt{2}} \frac{1}{\sqrt{2}} = \frac{e^{-4t}}{6}$

$y(t) = \frac{e^{-4t}}{3\sqrt{2}} \cos(2t - \frac{\pi}{4}) u(t) - \frac{e^{-4t}}{6} u(t)$

联系方式: _____

5-1] $\frac{d}{dt} = (9000) \cdot (97) \cdot 4V(12)$



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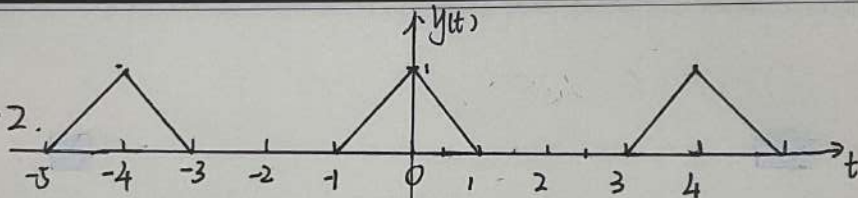
课程名称: _____

班级: _____ 教学班级: _____ 姓名: _____ 学号: _____ 第 _____ 页

2-23

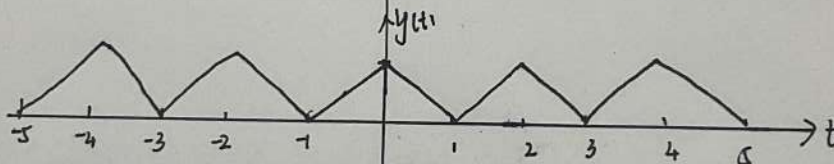
(a) $T=4$

$T > 2$

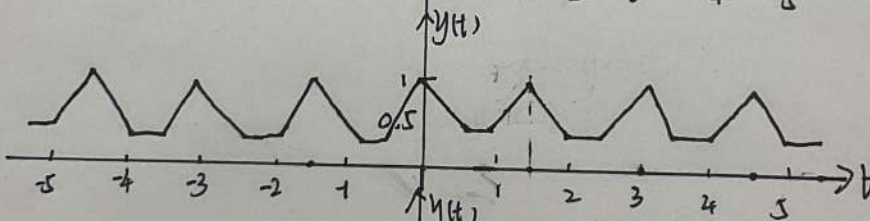


(b) $T=2$

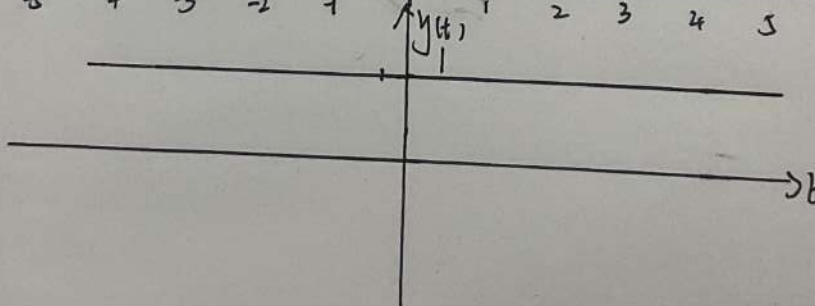
$T=2$



(c) $T=\frac{3}{2}$



(d) $T=1$



5-2-5